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SARDAR PATEL UNIVERSITY

M.Sc(IT) Sem-IV EXAMINATION, 2016

WEDNESDAY, 30th MARCH

10:30 am to 01:30 pm

COURSE CODE: (PS04FIIT01)

TITLE: Computer Based Numerical and Statistical Method

Maximum Marks: 70

- Q-1 Write the correct option in the answer book. [10]
- (1) From the following _____ method is the best method to obtain root of equation $f(x) = 0$.
(a) False position (b) Bisection
(c) Newton's Raphson (d) none of them
 - (2) True error is defined as
(a) Present Approximation - Previous Approximation
(b) True Value - Approximate Value
(c) $\text{abs}(\text{True Value} - \text{Approximate Value})$
(d) $\text{abs}(\text{Present Approximation} - \text{Previous Approximation})$
 - (3) The number 0.01850×10^3 has _____ significant digits
(a) 3 (b) 4 (c) 5 (d) 6
 - (4) To estimate the value of dependent variable x for a given value of independent variable y , the process is known as _____.
(a) Extrapolation (b) Inverse Interpolation
(c) Interpolation (d) polynomial
 - (5) _____ is called the backward difference operator.
(a) Δ (b) ∇ (c) $\emptyset$ (d) u
 - (6) _____ is not a type of interpolation method.
(a) Forward difference (b) backward difference
(c) Newton's divided difference (d) moving average method
 - (7) The system of linear equation $AX = B$ is said to be non-homogeneous if...
(a) $B \neq 0$ (b) $B = 0$ (c) $A \neq 0$ (d) A is symmetric
 - (8) Rate of change of distance with respect to time represents _____.
(a) Acceleration (b) Speed (c) Pressure (d) None of these
 - (9) Gradual, long-term movement in time-series data is called _____.
(a) seasonal variation (b) cycles (c) trends (d) exponential variation
 - (10) Cyclic variations in a time series are caused by:
(a) lockouts in a factory (b) war in a country
(c) floods in the states (d) none of the above
- Q:2 Answer the following in short. (Any Ten) [20]
- (1) Define Relative error and absolute error.
 - (2) Use the False Position method to obtain initial solution of $x^3 - 9x + 1 = 0$ (PTO)

- (3) Use the secant method to obtain initial solution of the equation $x^3 - 5x + 3 = 0$.
- (4) If x lies in the upper half of the table and if $x = x_k$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?
- (5) Define Interpolation.
- (6) Write algorithm for Backward difference table
- (7) Write different methods of Interpolation.
- (8) Explain central difference method.
- (9) Solve the system of equations (Using matrix inversion method)
 $10x + 3y = 5$, $8x - 2y = 2$.
- (10) Write a note on survey method of forecasting.
- (11) List the component of Time series.
- (12) Write merits and demerits of 'Ratio to Moving Average' method.

Q-3

- (a) Write algorithm for Regular Falsi method. [05]
- (b) Find the root of equation $x^3 - 4x - 9 = 0$ correct up to four decimal places using bisection method [05]

OR

Q-3

- (c) Find the root of equation $2^x - x - 3 = 0$ correct up to four decimal places using Iterative method. (Take $a = -3$). [05]
- (d) Find the root of equation $x^3 - 2x - 5 = 0$ correct up to four decimal places using Secant method. [05]

Q-4

- (a) Given the table of values as (use divided difference method) [05]

x	2.5	3.0	3.5	4.75	6.0
y(x)	8.85	11.45	20.66	22.85	38.60

Find $y(2.35)$.

- (b) The following table gives the census population of a town for the years 1931 to 1971. Estimate the population for the year 1965 by using an appropriate interpolation formula. [05]

Year	1931	1941	1951	1961	1971
Population	46	66	81	93	101

OR

Q-4

- (c) Estimate by Newton's method of interpolation, the expectation of life at age 32 from the following table. [05]

Age (In Years)	10	15	20	25	30	35
Expectation of life (In Year)	35.3	32.4	29.2	26.1	23.2	20.5

- (d) Given a function in the form of a table as [05]

x	2.0	3.0	4.0
y(x)	6.6	9.2	8.6

Interpolate the value of y(x) using Lagrangian polynomial at x= 2.8

Q-5

- (a) Solve the following system of equations using matrix inversion method. [05]

$$2x_1 + 8x_2 + 2x_3 = 14$$

$$x_1 + 6x_2 - x_3 = 13$$

$$2x_1 - x_2 + 2x_3 = 5$$

- (b) Write the comparison between direct and iterative methods for solution of system of linear equations. [05]

OR

Q-5

- (c) The distance (s) covered by a car in given time (t) is given in the following table [05]

Time(Miutes)	10	12	14	16	18
Distance(km)	12	15	20	27	37

Determine the acceleration of the car at t = 13 minutes.

- (d) Given the following table [05]

x	10	11	12	13	14
y	15	12.8	10.6	8.5	6.4

Estimate the value of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at 10, 10.5, 11, and 11.2 using the forward difference formula.

- Q.6 Obtain the trend from the time series given below by method of moving average of [i] 3 years [ii] 5 years [06]
(a)

Year	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970
Y	500	540	550	530	520	560	600	640	620	640

- (b) Explain steps of 'Ratio to moving Average' Method. [04]

OR

- Q.6 Calculate the seasonal index number for each of the four quarters for [06]
(a) following data.

Year	1 st Quarter	2 nd Quarter	3 rd Quarter	4 th Quarter
1980	106	124	104	90
1981	84	114	107	88
1982	90	112	101	85
1983	76	94	91	76
1984	80	104	95	83
1985	104	112	102	84

- (b) Explain Extrapolation model in forecasting. [04]

$$\bar{X} = \bar{X} = \bar{X}$$